UESTC4003 - Control

MATLAB Based Design Exercise:
Hybrid Energy Storage System for Grid Stability

Name: Chen Yichi
UESTC ID: [redacted] GUID[redacted]

December 5, 2025

Abstract

This report investigates the design and simulation of a Hybrid Energy Storage System (HESS) integrating a battery bank and a super-capacitor. The primary objective is to mitigate power fluctuations caused by wind power penetration and maintain grid stability. Using MATLAB Simulink, a frequency-based control strategy was developed and optimized. The system was rigorously tested under constant load and transient disturbance scenarios. Simulation results demonstrate that the proposed HESS effectively utilizes the high energy density of batteries for long-term balancing and the high power density of super-capacitors for fast transient response.

1 Introduction

Global energy trends are shifting towards renewable sources to reduce carbon emissions. However, the increasing integration of wind power into national grids poses significant challenges to power quality and stability due to its inherent intermittency and stochastic nature [1]. Unlike traditional synchronous generators, wind turbines typically lack rotational inertia, making the grid vulnerable to frequency deviations during sudden load changes or wind gusts [2].

To address these issues, Energy Storage Systems (ESS) are deployed to act as buffers. A single storage technology often fails to meet the conflicting requirements of high energy capacity (for long-term shifting) and high power capability (for transient stabilization). Therefore, a Hybrid Energy Storage System (HESS) combining complementary technologies is a promising solution [3]. This report designs and analyzes a HESS model to control power generation output and satisfy transient load demands using a frequency-separation control strategy.

2 Theoretical Background

2.1 Storage Technology Comparison

The HESS leverages the strengths of two distinct storage devices:

- **Battery Bank (High Energy):** Batteries, such as Li-ion or Lead-acid, offer high energy density, making them suitable for time-shifting large amounts of energy over

hours. However, they suffer from slow dynamic response and reduced cycle life when subjected to high-frequency current ripples [4].

- **Super-capacitor (High Power):** Super-capacitors (SCs) possess low energy density but extremely high power density and rapid charge/discharge rates. They can handle millions of cycles without degradation, making them ideal for absorbing high-frequency power surges and transient disturbances [5].

By combining these two, the battery handles the "base load" variation, while the SC acts as a "fast filter," protecting the battery from stress.

3 System Modeling and Design

3.1 System Architecture

The modeled microgrid consists of four main components: a wind generator, a battery bank, a super-capacitor, and a variable load. The system is governed by the instantaneous power balance equation:

$$P_{load}(t) = P_{wind}(t) + P_{battery}(t) + P_{SC}(t) \tag{1}$$

where $P_{battery}$ and P_{SC} are positive during discharging and negative during charging.

3.2 Wind Data and Source Modeling

To ensure realistic simulation results, real-world wind power data ("gridwatch_wind_data.csv") was used as the input source.
Modeling Approach: Instead of modeling the aerodynamic physics of a wind turbine (which requires complex wind speed-to-power conversion), a data-driven approach was adopted. The wind profile is modeled as a time-variant power source using the 'From Spreadsheet' block in Simulink. As shown in Fig. 1, the wind power fluctuates significantly, often exceeding the base load of 450 kW, necessitating a robust storage solution.



Figure 1: Wind power input profile over 24 hours.

3.3 Control Strategy: Frequency Separation

The core of the system is the control logic (Fig. 2). A classical Low-Pass Filter (LPF) is employed to allocate power based on frequency:

$$P_{total_deficit} = P_{load} - P_{wind} \tag{2}$$

$$P_{bat_ref} = \frac{1}{1+\tau s} \cdot P_{total_deficit} \tag{3}$$

$$P_{SC_ref} = P_{total_deficit} - P_{bat_ref} \tag{4}$$

The time constant τ is the critical tuning parameter. A larger τ directs more high-frequency components to the super-capacitor, resulting in a smoother battery profile.

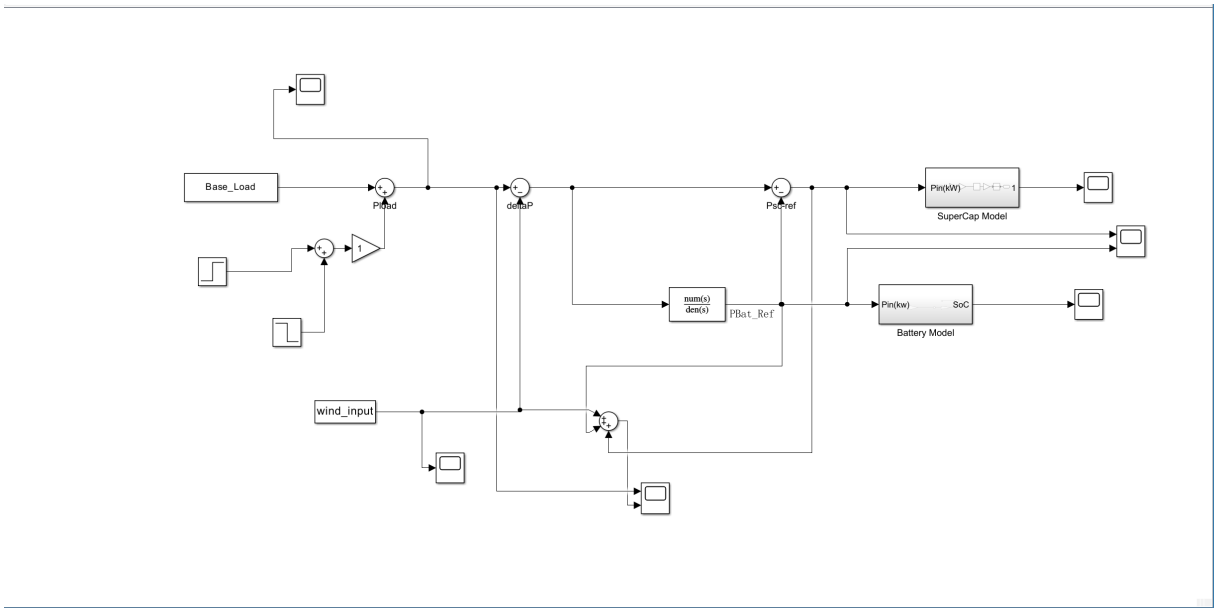


Figure 2: Overall Simulink Block Diagram implementing the LPF control strategy.

3.4 Subsystem Implementation

Detailed subsystem designs are shown in Fig. 3.

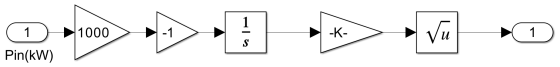
- **Battery Model:** Implements the Coulomb Counting method. It integrates the power (approximated as current) to calculate the State of Charge (SoC):

$$SoC(t) = SoC_{init} - \frac{1}{E_{cap}} \int P_{bat}(t)dt \tag{5}$$

- **Super-capacitor Model:** Monitors the energy state to ensure it responds to rapid charging/discharging commands [6].



(a) Battery Subsystem



(b) Super-capacitor Subsystem

Figure 3: Detailed Subsystem Modeling in Simulink.

3.5 Key Parameters

The final optimized parameters used in the simulation are listed in Table 1.

Table 1: Key Simulation Parameters of the HESS.

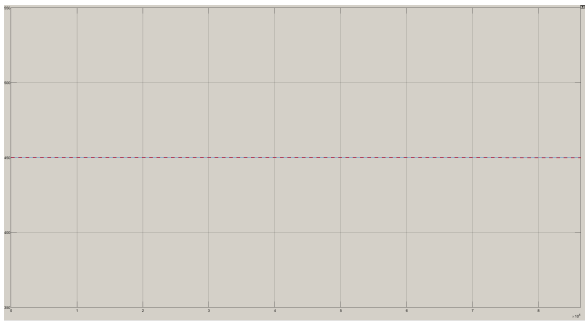
Component	Parameter	Value
Wind & Load	Base Load	450 kW
	Load Disturbance Step	+300 kW
Battery Bank	Initial SoC	50 %
	Optimized Capacity	**6000 kWh**
Super-capacitor	Capacitance	500 F
	Rated Voltage	270 V
	Initial Voltage	270 V
Controller	Time Constant (τ)	**300 s** (Optimized)

4 Simulation Results and Analysis

4.1 Case 1: Constant Load Performance

The system was tested with a constant load of 450 kW. The goal was to optimize battery capacity using a trial-and-error method.

Power Balance and Stability: Fig. 4a confirms that the total supply matches the load demand. Even with a constant load, the super-capacitor is active (Fig. 4b), filtering high-frequency wind turbulence.



(a) Power Balance



(b) SC Filtering

Figure 4: Case 1 Performance Analysis.

Battery SoC Optimization (Trial and Error): Initially, the battery capacity was set to 3000 kWh. Simulation results showed that due to significant surplus wind energy, the SoC exceeded the 100% limit, reaching nearly **140%** by the end of the day.
By incrementally increasing the parameter, a final capacity of **6000 kWh** was selected. With this optimized value, the SoC curve (Fig. 5) stabilizes around **90%**, ensuring safe operation.

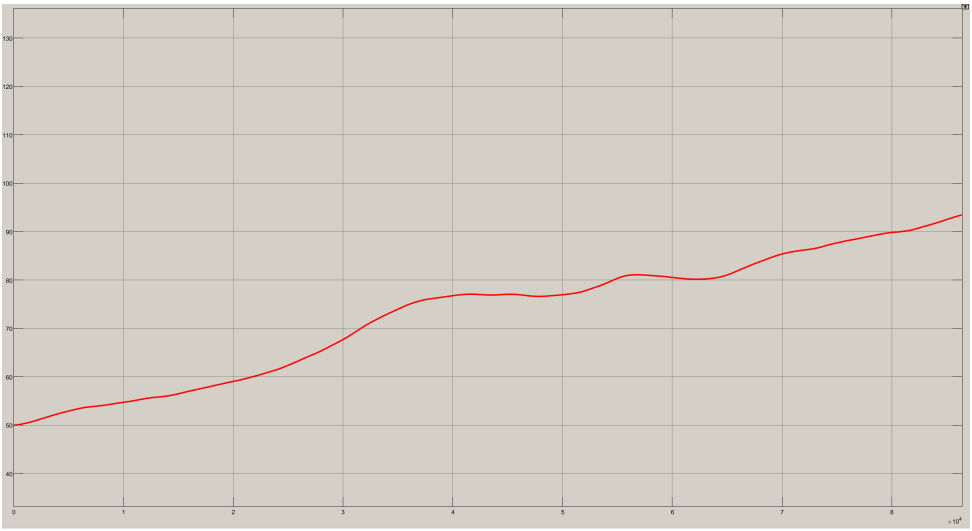


Figure 5: Case 1 Battery SoC: Stabilized curve with 6000 kWh capacity.

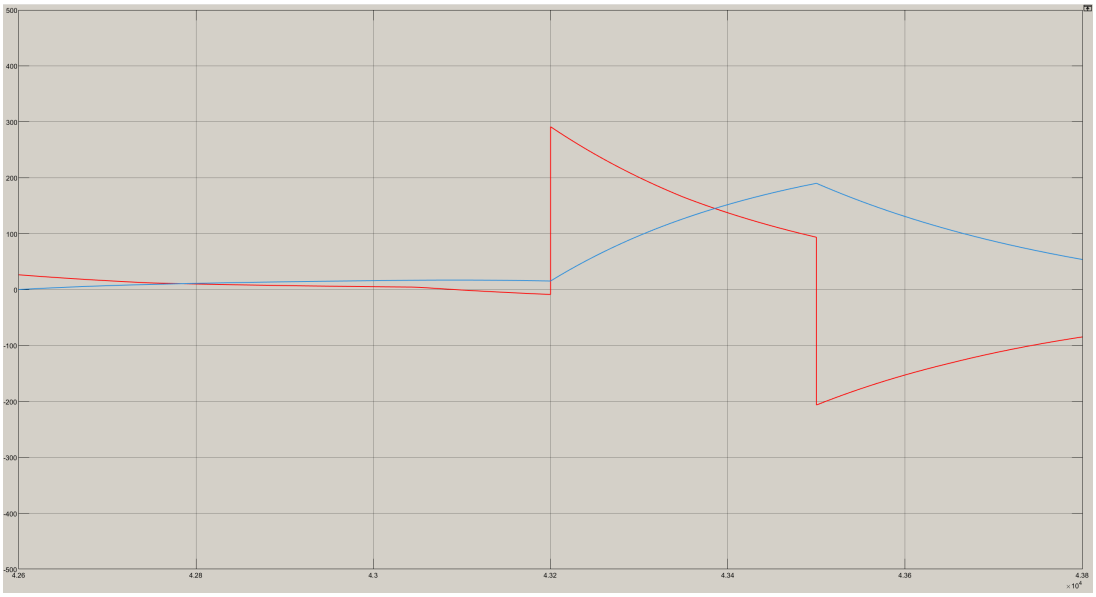
4.2 Case 2: Load Disturbance Response

A disturbance was introduced at 12:00 PM (load step to 750 kW). The objective was to optimize the time constant τ.

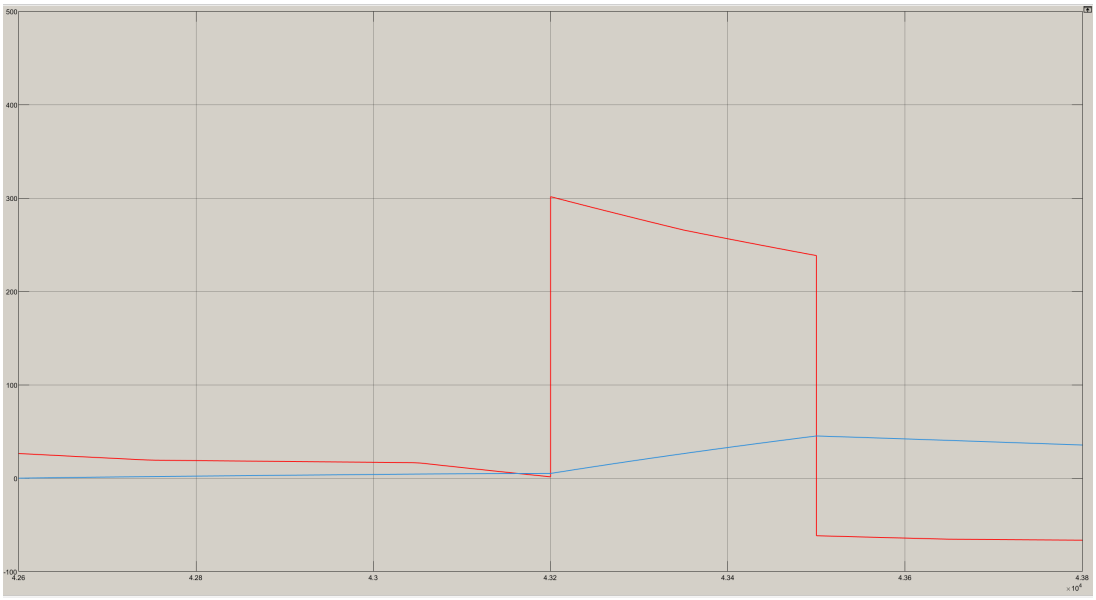
Optimization Analysis ($\tau = 300s$ vs. $\tau = 2000s$): Comparing different time constants yielded insights into system recovery (Fig. 6):

- **Initial Response:** For both settings, the super-capacitor (P_{SC}) spikes to **300 kW** to meet the demand.
- **Recovery Capability:**
  - With $\tau = 300s$ (Fig. 6a), P_{SC} drops to -200 kW after the load is removed and then **slowly returns to 0**. This V-shape indicates the battery successfully took over, allowing the SC to reset.
  - With $\tau = 2000s$ (Fig. 6b), the dip is smaller (-70 kW), but P_{SC} shows **almost no trend of returning to 0**, meaning it remains stressed.

Therefore, $\tau = 300s$ was selected as the optimal value.



(a) Optimized Response ($\tau = 300s$): P_{SC} returns to zero.



(b) Slow Response ($\tau = 2000s$): P_{SC} fails to reset.

Figure 6: Case 2 Transient Response Comparison.

5 Conclusion

The simulation confirms that the hybrid system effectively maintains grid stability.

1. In Case 1, increasing the battery capacity from 3000 kWh to 6000 kWh prevented overcharging (140% to 90%).

2. In Case 2, an optimized time constant ($\tau = 300s$) allows the super-capacitor to handle spikes and recover effectively, whereas a larger time constant ($\tau = 2000s$) fails to reset the system state.

References

[1] X. Li, D. Hui, and X. Lai, "Review of energy storage system for wind power integration support," *Appl. Energy*, vol. 104, pp. 137–147, 2013.

[2] L. Gelažanskas, A. Baranauskas, K. A. A. Gamage, and M. Ažubalis, "Hybrid wind power balance control strategy using thermal power, hydro power and flow batteries," *Int. J. Electr. Power Energy Syst.*, vol. 74, pp. 310–321, 2016.

[3] H. Zhou, T. Bhattacharya, D. Tran, T. S. T. Siew, and A. M. Khambadkone, "Composite energy storage system involving battery and supercapacitor in a microgrid application," *IEEE Trans. Power Electron.*, vol. 26, no. 3, pp. 923–930, 2011.

[4] M. Khalid and A. V. Savkin, "Improving the energy efficiency of a battery/supercapacitor hybrid energy storage system," *IEEE Trans. Veh. Technol.*, vol. 59, no. 3, pp. 1127–1132, 2010.

[5] L. Zubieta and R. Bonert, "Characterization of Double-Layer Capacitors for Power Electronics Applications," *IEEE Trans. Ind. Appl.*, vol. 36, no. 1, pp. 199–205, 2000.

[6] MATLAB, "Supercapacitor Parameter Identification," 2018. [Online]. Available: https://uk.mathworks.com/help/physmod/sps/examples/supercapacitor-parameter-identification.html.